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MILITARY SPECIFICATION

CHILLER UNITS, LIQUID, PACKAGED, ELECTRIC MOTOR-DRIVEN WITH RECIPROCATING COMPRESSORS 10 TONS TO 150 TONS

1. SCOPE

1.1 Scope. This specification covers packaged liquid chiller units with electric motor-driven reciprocating compressors. The units serve as a source of chilled liquid for air conditioning and refrigerated systems.

1.2 Classification. The chiller units covered by this specification shall be of the following types, styles, and sizes as specified (see 6.2).

Type I - Chiller units with integral condenser.

Size - 10 tons to 80 tons.

Type II - Chiller units for use with remote condenser.

Size - 10 tons to 150 tons.

Style A - Water-cooled condenser.
Style B - Air-cooled condenser.
Style C - Evaporative condenser.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-V-173	Varnish, Moisture-and-Fungus Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).

FSC 4130

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MIL-F-3541 Fittings, Lubrication.
MIL-R-3593 Refrigeration and Cooling Equipment (Excluding Household Refrigerators), Packaging of.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 Quality Assurance Terms and Definitions.
MIL-STD-129 Marking for Shipment and Storage.
MIL-STD-130 Identification Marking of U.S. Military Property.
MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System.
MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Air Conditioning and Refrigeration Institute (ARI)

ARI Standard 450 - Standard for Water-Cooled Refrigerant Condensers, Remote Type.
ARI Standard 460 - Standard for Remote Mechanical-Draft Air-Cooled Refrigerant Condensers.
ARI Standard 495 - Standard for Refrigerant Liquid Receivers.
ARI Standard 590 - Standard for Reciprocating Water-Chilling Packages.

(Application for copies should be addressed to the Air Conditioning and Refrigeration Institute, 1815 North Fort Myer Drive, Arlington, VA 22209.)

American National Standards Institute (ANSI)

ANSI B9.1 - Safety Code for Mechanical Refrigeration.
ANSI C1 - National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc. (ASHRAE)

ASHRAE Standard 30 - Liquid Chilling Packages, Methods of Testing for Rating.
ASHRAE 34 - Refrigerant, Number Designation of.

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., United Engineering Center, 345 East 47th Street, New York, NY 10017.)

American Society of Mechanical Engineers (ASME)

ASME Rules for the Construction of Unfired Pressure Vessels (Section VIII).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

American Society for Testing and Materials (ASTM)

ASTM B209 - Aluminum-Alloy Sheet and Plate.
ASTM B210 - Aluminum-Alloy Drawn Seamless Tubes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Welding Society (AWS)

Codes for Welding.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

National Bureau of Standards (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

National Electrical Manufacturers Association (NEMA)

NEMA MG-1 - Standards for Motors and Generators.
NEMA ICS - Standards for Industrial Control.

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(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

Underwriters' Laboratories, Inc. (UL)

UL207(c) - Standard for Safety Refrigerant-Containing Components.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The liquid chiller (hereinafter called "chiller unit"), covered by this specification shall consist of a complete packaged liquid chilling plant with water cooler, compressor, condenser, operating and safety controls including a flow switch in the chilled water line, refrigerant piping, and control piping. The entire unit shall be factory-assembled and rated in conformance with ARI 590, and shall be provided with all the standard equipment and accessories listed therein, and ready for operation after being charged and connected to the appropriate utilities. Heat exchangers and liquid receivers shall be provided on refrigerant units when the system refrigerant charge exceeds 80 percent of the condenser refrigerant volume. Chiller units consisting of two or more compressors, but not more than four, may be provided when mounted on a single base. Where compressors are paralleled, not more than two shall be connected and not less than two independent refrigerant circuits shall be provided. Chiller capacity shall be based on conditions specified herein. The overall dimensions (length, width, and height) of the units shall be those normally supplied by the manufacturer, except that when limitations are required, the dimensions shall be as specified (see 6.2).

3.2 Safety standards. The chiller unit shall conform to the applicable requirements of ANSI B9.1. Refrigerant-containing components shall be in accordance with UL207(c) or ASME Section VIII.

3.3 Standard product. Unless otherwise specified herein, the chiller unit shall be the standard product of the manufacturer and shall be identifiable by the manufacturer's model number, figure number, or equivalent designation. The chiller unit shall be new and unused except to the extent required to perform the tests specified herein.

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3.4 First article. When specified (see 6.2), the contractor shall furnish a chiller unit for first article inspection and approval (see 4.3 and 6.3).

3.5 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality normally used by manufacturers in commercial practice. All material shall be modern in design and shall not have been in prior service except as required by factory tests. Items performing the same functions shall be products of the same manufacturer. All weld areas and deposited weld metal shall be equally as corrosion-resistant as the parent material.

3.6 Design. The unit shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.7 Interchangeability. All chiller units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.8 Refrigerant. Refrigerant shall conform to ASHRAE 34.

3.9 Electrical wiring. Electrical wiring shall be installed in accordance with ANSI C1 for use with the electrical power specified (see 6.2).

3.10 Motors. Motors for the compressor shall be of the open enclosure or hermetically sealed type (refrigerant-cooled windings) and shall conform to NEMA MG-1. All motors, including those used on the condensers, shall be designed to operate on the electric service specified in 3.9.

3.10.1 Motor controllers. Motor control equipment shall conform to NEMA ICS. Sealed motor controllers shall be underwriter approved. Overload protective devices shall be provided for motor protection. Starters shall be across-line, part wind, wye delta, or automatic transformer design of the closed transition type as specified (see 6.2).

3.11 Compressor. The compressor shall be a heavy duty open or hermetically sealed, as specified (see 6.2), reciprocating, constant speed type, equipped with forced feed lubrication, crankcase heater, and suction strainer. Each cylinder-unloading device shall be unloaded when the compressor starts. Piston speed for the compressor shall not exceed the recommendations of the manufacturer or 875 feet per minute (f.p.m.), whichever is less. The compressor shall operate at partial-load conditions without increased vibration at full load operation and shall be capable of continuous operation down to the lowest step of unloading. When specified (see 6.2) unit

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shall be provided with hot-gas muffler to reduce vibration and noise from pulsation.

3.12 Receiver. Receivers furnished separately shall conform to the applicable sections of ARI Standard 495.

3.13 Liquid cooler. The liquid cooler shall be direct expansion type or flooded type. The cooler shall be designed, fitted, and rated in accordance with ARI 590. The liquid cooler shall be constructed in accordance with the applicable portions of ASME Boiler and Pressure Vessel Code, Section VIII. Tubes shall be seamless copper. For water side of liquid cooler, performance shall be based on a water velocity of not less than 3 feet per second (f.p.s.) nor more than 10 f.p.s., and a fouling factor of 0.0005 or 0.001 as specified (see 6.2).

3.13.1 Direct expansion type. On direct expansion type units, each refrigerant circuit shall be complete with water solenoid valve and expansion device, capable of modulating the minimum step of capacity unloading. The water pressure drop through the water cooler shall be not greater than the pressure drop specified (see 6.2).

3.13.1.1 Expansion device. For direct expansion systems, an expansion device shall be installed in the refrigerant line adjacent to the inlet of each liquid cooler. The remote bulb of the expansion valve shall be either liquid or vapor charged, securely attached to or inserted in the refrigerant suction line. A liquid line strainer shall be installed when an expansion device is furnished.

3.13.2 Flooded type. For flooded type chiller units, an expansion device shall be located between the water outlet of the condenser and the liquid cooler inlet to maintain a proper refrigerant level in the liquid cooler and to prevent gas bypass. The valve shall be readily accessible for inspection and maintenance, and shall be of corrosion-resistant metal.

3.13.3 Heat exchanger or economizer. The heat exchanger or economizer, when required by unit, shall be the manufacturer's standard design and suction-liquid-line type that subcools the liquid and super-heats the gas, preventing liquid-line slopover and increasing unit capacity. Heat exchangers shall be constructed and tested in accordance with ANSI B9.1.

3.14 Type I - Chiller unit with integral condenser. Type I chiller units shall have the condenser unit an integral part of the chiller unit. Capacities shall be from 10 to 80 tons, as specified in 1.2.

3.15 Type II - Chiller unit with remote condenser. Type II Chiller units shall have remote condensers. Capacities shall be from 10 to 150 tons, as specified in 1.2.

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3.16 Style A - Water-cooled condensers. Water-cooled condensers shall be of the shell- and coil-type or shell- and tube-type. Remote type condensers shall be designed, fitted, and rated in accordance with ARI 450, and shall be provided with all the standard equipment and accessories listed therein. Condensers shall be constructed and tested in accordance with the applicable portions of section VIII of the ASME Boiler and Pressure Vessel Code. For the design conditions stated on the manufacturers drawing, the contractor shall furnish a cross-plotting of the net refrigeration effect (NRE) of the condenser against the NRE of the compressor to provide the NRE indicated on the drawing.

(a) Shell and tube condenser shall have seamless or welded steel shell, steel tube sheets, and cast-iron or steel water boxes. Tubes shall be nonferrous metal, plain or integral finned, and shall be expanded full diameter into reamed and grooved holes or silver soldered. Intermediate tube supports shall be provided so that lengths of straight tubing between supports will not exceed 3 feet for copper tubes and 4 feet for brass tubes. Tubes shall fit tightly in the supports to prevent chafing due to vibration or pulsations.

(b) Shell and coil condenser shall have seamless or welded steel shell and welded head. Water shall flow through the coil, which shall be nonferrous metal, plain or integral finned, and arranged so that it can be drained completely. Coil joints shall be brazed or silver soldered. The entire bundle shall be removable.

3.16.1 Condenser controls. On a decrease in refrigerant discharge pressure, a pressure controller shall modulate a three-way valve to mix leaving condenser water. Control shall be set for a saturated refrigerant condensing temperature of 105° Fahrenheit (F.).

3.16.2 Water cooling section. The water-cooling sections shall be of nonferrous metal, an approved nonmetallic material, or of not lighter than 24-gage, 0.0276 inch in nominal thickness, steel hot-dip galvanized after fabrication. Spray nozzles shall be brass non-clogging type designed to permit easy disassembly, and shall be arranged for easy access. The water pump shall be a bronze-fitted centrifugal or turbine type and may be mounted as an integral part of evaporative condenser or remotely on a separate mounting pad. Pump-motors shall be totally enclosed. Motor starters shall be magnetic across-the-line type with watertight enclosure. Pump suction shall be fully submerged and provided with screened inlet. Water pan or sump shall be not lighter than 14-gage, 0.0785 inch in nominal thickness, steel hot-dip galvanized after fabrication or molded acid-resistant fiberglass reinforced polyester. Water distribution pan shall be not lighter than 16 gage, 0.0635 inch in nominal thickness, steel hot-dip galvanized after fabrication. Joints shall be watertight. Water pan or sump shall

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be provided with drain and overflow. Make up water connection shall be provided with stop valve and float valve. A bleed line with a flow valve or fixed orifice in pump discharge line shall be provided.

3.17 Style B - Air-cooled condensers. Air-cooled condensers shall be suitable for remote installation in a weather-protected casing or as an integral part of condensing unit. The air-cooled condenser shall be a complete factory-fabricated and assembled unit consisting of coils, fans, and electric-motor drive. Capacity at design conditions shall be based on 30 degrees F. temperature differential between entering air and condensing refrigerant. The saturated refrigerant condensing temperature shall be not over 130° F. The entering dry bulb outside air temperature shall be based on 90° F. For the design conditions stated on the manufacturer's drawings, the contractor shall furnish a cross-plotting of the net refrigeration effect (NRE) of the condenser against the NRE of the compressor to provide the NRE indicated on the drawings and to show the compatibility of the equipment furnished. Sub-cooling shall not be taken into account in determining compressor and condenser capacities. The air-cooled condenser may be used for refrigerant storages in lieu of a separate receiver, provided that condenser storage capacity is 20 percent in excess of fully charged system. Remote type air cooled condensers shall be tested and rated in accordance with ARI 460.

3.17.1 Condensing coils for multi-compressors. For multi-compressors, a separate air-cooled condenser for each compressor in multi-compressor installations shall be provided. Separate condensing coils or circuits in a common housing may be used, provided each coil is provided with separate controls to operate individual fans for each coil. All coils shall have sub-cooler and receiver.

3.17.2 Condenser coil. Condenser coil shall be extended-surface fin-and-tube type and shall be seamless copper or aluminum tubes with copper or aluminum fins. Condenser coils for installation in salt air or corrosive atmosphere shall be constructed of aluminum tubes, fins and end sheets; copper tubes, fins and end sheets; or copper tubes and fins and galvanized-steel end sheets. On condensers with all aluminum construction, aluminum alloy conforming to ASTM B210, alloy 1100, shall be used for the tubes, and aluminum alloy conforming to chemical requirements of ASTM B209, alloy 7072, shall be used for fins and sheets. Fins shall be soldered or mechanically bonded to tubes and installed in a metal casing. Coil shall be tested after assembly at pressures specified in ANSI B9.1 for refrigerant employed in system. After testing, coil shall be dried to remove free moisture and capped to prevent entrance of foreign matter. Coil shall be evacuated and sealed at factory.

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3.17.3 Fans. Fans shall be either centrifugal or propeller driven as best suited for application. Fans shall be directly connected to motor shaft, or indirectly connected to motor by means of V-belt drive. Adjustable sheaves shall be provided. Fans shall be statically and dynamically balanced.

3.17.4 Electric motor. Electric motor shall conform to NEMA MG-1, and shall be totally enclosed type, or open type so located within an enclosure as to be fully protected from weather. Motor starter shall be magnetic across-the-line type with weather-resistant enclosure. Thermal protection shall be manual or automatic reset type.

3.17.5 Condenser casing. Condenser casing shall be aluminum of not less than 0.040-inch nominal thickness or hot dipped galvanized steel of not less than 18 gage 0.0516 inch in nominal thickness. Condenser shall have horizontal air discharge with a discharge baffle to direct air upward, and shall be constructed of same material and thickness as casing. Wire screens shall be installed over all fans, motors,and drives that are not adequately protected by housing.

3.17.6 Discharge air. Discharge air in each air cooled condenser shall be in vertical direction either directly from fan casing or by means of supplementary wind deflectors.

3.17.7 Condenser controls. Head pressure control shall be provided to insure condensing temperature for proper system operation at all ambient temperatures down to 0° F.

3.17.7.1 Air volume control. Air volume control will not be permitted in areas where the expected ambient temperature is below 35° F. On a decrease in refrigerant discharge pressure, volume-control dampers shall modulate to control airflow over condenser coil. Solid state variable speed fan motor controller may be provided in lieu of volume dampers to control air flow over coil. Condensers with multiple fans may be provided with fan cycling control to cycle one of two fans, two of three fans, or four of six fans in response to outdoor ambient temperature. Control shall be set for minimum of 95° F. saturated refrigerant condensing temperature.

3.17.7.2 Condenser flooding controller. Condenser flooding controller shall be provided to throttle liquid flow from condenser outlet and thereby increase the amount of liquid in the condenser. A separate receiver for storage of excess refrigerant shall be provided. Controls shall be set for a minimum of 95° F. saturated refrigerant condensing temperature.

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3.18 Style C - Evaporative condenser. Evaporative condensers shall be complete with fans, water pump with suction strainer, electric motor and drive equipment, water eliminators if required, condensing coil, liquid sub-cooler, liquid receiver if required, water pan or sump, spray nozzles or water-distribution pan, water strainer, water makeup assembly, bleeder with flow valve or a fixed orifice, enclosure with suitable access doors, and air inlet and outlet openings. No water shall carry over into discharge outlet of the unit. For the design conditions shown on the manufacturer's drawings, the contractor shall furnish a cross-plotting of the NRE of the evaporative condenser against the NRE of the compressor to provide the NRE indicated on the drawings and to show the compatibility of the equipment furnished.

3.18.1 Liquid subcooler. Each evaporative condenser, other than close coupled systems, shall be provided with liquid sub-cooler to reduce temperature to not less than 10° F. below condensing temperature. Sub-cooling shall not be taken into account in determining compressor and condenser capacities. A receiver shall be furnished and piped into the system, so that a liquid seal is obtained in liquid line leaving condenser.

3.18.2 Refrigerant section. Condenser coil shall be unfinned copper or steel tubes hot-dip galvanized after fabrication. Receiver shall be welded steel and shall be fitted and tested in conformity with ARI 495. A refrigerant charging valve shall be installed in liquid line between receiver cutoff valve and expansion device. The refrigerant section shall be tested at pressure specified in ANSI B9.1 for refrigerant placed in system. After testing, refrigerant circuit shall be dried to remove free moisture and capped to prevent entrance of foreign matter.

3.18.3 Fans. Fans shall be hot-dip galvanized steel or stainless steel, centrifugal or propeller type, connected through an adjustable sheave to an electric motor. Electric motor shall be totally enclosed type or open dripproof so located within an enclosure as to be fully protected from weather. Motor starters shall be magnetic across-the-line type with weather-resistant enclosures.

3.18.4 Water section. Water eliminators shall be nonferrous metal, an approved nonmetallic material, or of not lighter than 24-gage, 0.0276 inch in nominal thickness, steel hot-dip galvanized after fabrication. Spray nozzles shall be brass non-clogging type designed to permit easy disassembly, and shall be arranged for easy access. Water pump shall be bronze-fitted centrifugal or turbine type and may be mounted as an integral part of evaporative condenser or remotely on a separate mounting pad. Pump-motors shall be totally enclosed. Motor starters shall be magnetic across-the-line type with weather-resistant enclosure. Pump suction shall be fully submerged

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and provided with screened inlet. Water pan or sump shall be not lighter than 14-gage, 0.0785 inch in nominal thickness, steel hot-dip galvanized after fabrication, or molded acid-resistant fiberglass reinforced polyester. Water distribution pan shall be not lighter than 16 gage, 0.0635 inch in nominal thickness, steel hot-dip galvanized after fabrication. Joints shall be watertight. A water pan or sump with drain, overflow, and makeup water connection with stop valve and float valve shall be provided. A bleed line with a flow valve or fixed orifice in pump discharge line shall be provided.

3.18.5 Enclosure. Enclosure shall be not lighter than 18-gage hot-dip galvanized steel, 0.0516-inch in nominal thickness, and adequately reinforced and braced. Access doors or panels suitably sized and located shall be provided for access to water nozzles or distribution pan, coils, and valves for cleaning, repair, or removal of item. Access doors or panels shall be gasketed with synthetic rubber, or other approved gasket material, and locked in place with thumb screw or catches. Angles, frames, or flanges for all duct connections shall be provided . One-half-inch mesh hot-dip galvanized steel or copper air-inlet screens shall be installed on each air-inlet.

3.18.6 Head pressure control. On a decrease in refrigerant discharge pressure, fan discharge or inlet air dampers shall modulate to reduce air flow through evaporative condenser, or a solid state variable speed fan motor controller may be provided in lieu of volume dampers to control air flow. On a decrease in refrigerant discharge pressure, discharge, bypass, and air-inlet dampers shall modulate to bypass the evaporative condenser and to discharge air into the evaporative condenser air inlet. Control shall be set for a saturated refrigerant condensing temperature of 105° F.

3.19 Operating and safety controls. Water-chiller units shall be provided with operating and safety controls of the electric, electronic, or pneumatic type. Water chillers shall be provided with a factory-piped-and-wired control panel fitted with a discharge-pressure gage, suction pressure gage, oil pressure gage, separate high-pressure cutout with manual reset separate low-pressure cutout, oil-pressure differential switch, low-temperature cutout with manual reset, compressor operating control, manual off-automatic switch, operating pilot light, and any other relays or appurtenances necessary for safe controlled chiller operation. The controls shall be provided with devices or means to protect each compressor from short cycling whenever the compressor is shut-down by any of the safety controls. Pump down cycle shall be provided for each compressor 20 tons and over. Alarm circuit terminals shall be provided in control panel to actuate a 4-inch-diameter alarm bell in event of machine cutout on any protective device except when low-pressure control is used as an operating

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control. A transformer shall be provided for the bell circuit. System capacity control, suction or temperature actuated, shall be provided to adjust compressor capacity in accordance with system load and to automatically recycle the system on power interruption. The temperature sensing element shall be located in the chilled water return line to the chiller. For systems of from 10 to 51 tons, the number of steps of automatic capacity reduction shall be 2 steps, and for systems of 52 tons and over, capacity reduction shall be 4 steps. Capacity reduction may be provided by cylinder unloading, by sequence operation of two or more, but not to exceed four, compressors, or by a combination of the two methods.

NOTE: The desired number of steps of capacity reduction should be checked for availability.

3.20 Piping and accessories. All piping and accessories required for completion of the chiller unit shall be provided by the supplier. Piping, valves, fittings, and related parts shall conform to ANSI B9.1 and applicable portions of the ASME pressure vessel code.

3.21 Application conditions. A ton of refrigeration shall be equivalent to the extraction of 12,000 British thermal units (B.t.u.) per hour at the liquid cooler. Application data used in the selection of the equipment shall include the four items supplied by the procuring agency (see 6.2), the values for which shall appear in the manufacturer's bid proposal. Published ratings shall indicate the brake horsepower to the compressor shaft or kilowatt input to the compressor motor for those combinations of conditions specified in the ratings.

3.22 Lubrication. Unless otherwise specified (see 6.2), means for lubricating shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall be so located as to make the lubricating points easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery.

3.22.1 Grease fittings. Grease fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. A bright red circle shall be painted around each lubrication point.

3.23 Fungus resistance. When specified (see 6.2), electrical connections, including terminal and circuit connections, components, and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or hermetically sealed need not be treated.

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b. Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

3.24 Treatment and painting. Unless otherwise specified (see 6.2), the equipment shall be treated and painted in accordance with the manufacturer's standard practice.

3.25 Electromagnetic interference control. When specified (see 6.2), the equipment shall conform to the electromagnetic interference control requirements and test limits for class No. IIB equipment as specified in MIL-STD-461.

3.26 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The nomenclature of the item shall be Chiller Unit.

3.27 Spare parts and maintenance tools. When specified (see 6.2), spare parts and maintenance tools shall be furnished.

3.28 Workmanship. The quality of workmanship shall be such as to produce chiller units that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the unit.

3.28.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. Material shall be straightened by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Flame cutting, with use of a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing, provided that exposed edges are smoothly made. All bends of a major character shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating and heated metal shall be allowed to cool slowly.

3.28.2 Bolted connections. Bolt holes shall be accurately punched and reamed or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be completely fastened and tight. Bolts shall be fully engaged, but shall not extend more than two full threads.

3.28.3 Threads. Screw threads for all thread securing or attaching devices and threaded parts shall be in accordance with NBS H28 as follows:

a. Machine threads shall be the standard national coarse or fine thread series.

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b. Pipe threads shall be the standard national taper pipe thread series.

3.28.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be of the explosive type or driven with pressure tools, and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of a shape approved by the contracting officer and of uniform size for the diameter of the rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.5 Welding. The surfaces of parts to be welded shall be free of oxide, scale, paint, grease, or other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, or deformation of material. All scale or flux deposits (when flux is used) shall be removed from the finished welds. Weld penetration shall provide transference of maximum design stress through the base metaljuncture. Fillets shall be provided where necessary to reduce stress concentration. Welding shall be in accordance with an applicable code published or recommended by AWS.

3.28.6 Machine work. Tolerances for contact and bearing surfaces shall conform to standards prevailing among manufacturers normally producing chiller units.

3.28.7 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects that may render the casting unsound for use. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste shall not be attempted without authorization of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1,

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the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested to the extent specified, in accordance with the requirements of referenced subsidiary specifications and standards. The terms and definitions of MIL-STD-109 shall apply.

4.1.2 Acceptability criteria. Chiller units which conform to all requirements in Sections 3 and 5 of this specification and pass all examination and tests in Section 4 of this specification will be considered acceptable by the Government.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of preparation for delivery (see 4.9).

4.3 First article inspection. First article inspection shall be performed on one chiller unit when a first article sample is required (see 3.4). This inspection shall include the examination of 4.7 and tests of 4.8. The first article shall be a first production item.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on the sample chiller unit selected in accordance with 4.6.1. This inspection shall include the examination of 4.7 and the tests of 4.8.

4.5 Inspection lot. All units of the same type, style, and size offered to the Government at one time shall be considered a lot for purposes of inspection. The sample unit shall be one complete chiller unit.

4.6 Sampling.

4.6.1 Sampling for examination. A random sample of chiller units shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. In terms of defects per 100 units, the acceptable quality level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.6.2 Sampling for tests. A random sample of chiller units shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 4.0 percent defective.

4.7 Examination. The first article sample, when a first article sample is required, and samples selected in accordance with 4.6.1 shall be examined

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to verify compliance with this specification. Examination shall be conducted as specified in table I.

TABLE I. Classification of defects

Categories	Defects
Critical:	None defined.
Major:	
	101. Material not as specified.
	102. Like parts not interchangeable for chiller units of the same classification.
	103. Safety devices missing or not in accordance with ANSI B9.1.
	104. Compressor not reciprocating type.
	105. Condenser not equipped with pressure relief device.
	106. Float valve not accessible for inspection or maintenance.
	107. Water chilling system automatic controls not complete.
	108. Means for lubrication not provided, or in inaccessible locations (when lubrication is required).
	109. Grease fittings not as specified, or in inaccessible unprotected locations.
	110. Workmanship not as specified.
Minor:	
	201. Spare parts and maintenance tools not furnished (when required).
	202. Lubrication points not marked with a red circle.
	203. Identification marking not as specified, illegible, missing, or incomplete.

4.8 Tests. Each first article sample, when a first article sample is required, and each sample selected in accordance with 4.6.2 shall be tested, and any unit failing to pass the following tests, as applicable, shall be rejected. Tests shall be conducted as outlined in the referenced documents as herein specified.

4.8.1 Operating test. Each chiller unit shall be properly set up and adjusted in accordance with the manufacturer's instructions. The unit shall be operated for a period long enough to determine that all mechanical and electrical controls and parts function accurately, efficiently, and safely. Nonconformance to any of the requirements of Section 3 shall constitute failure of this test.

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4.8.2 Leak test. Each chiller unit shall be tested for tightness under pressure, and shall be tested by any method acceptable to the procuring activity, at not less than the minimum refrigerant leak field test pressure shown in ASHRAE Standard 30 (ANSI B9.1). Any leakage of refrigerant or oil shall constitute failure of this test.

4.8.3 Electromagnetic interference control. When conformance with electromagnetic interference limits is required, the equipment shall be tested in accordance with MIL-STD-462 to determine conformance with electromagnetic interference control requirements of 3.25.

4.9 Inspection of preparation for delivery. The preservation, packaging, packing, and marking specified in section 5 of this specification shall be examined and tested to determine compliance with the applicable quality assurance provision of MIL-R-3593.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Each chiller unit, accesories, spare parts, and tools shall be preserved, packaged, packed, and marked in accordance with MIL-R-3593. Preservation and packaging shall be level A or C as specified (see 6.2), and the level of packing shall be level A, B, or C as specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The chiller units are intended for use in supplying chilled liquid for refrigeration and air conditioning systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and size of chiller unit required (see 1.2).
(c) Dimensions of the unit required if space is limited (see 3.1).
(d) When a first article chiller unit is required (see 3.4).
(e) Electrical service on which the chiller units will operate (see 3.9).
(f) Type motor controller required (see 3.10.1).
(g) Whether the compressor shall be a heavy duty open type or hermetically sealed, and when muffler is required (see 3.11).
(h) Basis of performance for water side of liquid cooler (see 3.13).
(i) Maximum water pressure drop allowed through the water cooler (see 3.13.1).

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(j) Application conditions with values for the following (see 3.21):
    1. Quantity of chilled liquid in gallons per minute (g.p.m.).
    2. Temperatures of water entering and leaving the cooler in degrees F.
    3. Quantity of condenser water in g.p.m.
    4. Temperature of available condenser water in degrees F.
(k) When a means of lubrication is not required (see 3.22).
(l) When fungus resistance is requried (see 3.23).
(m) When special treatment and painting is required, and if so, the type of treatment and painting required (see 3.24).
(n) When electromagnetic interference control is required (see 3.25).
(o) Spare parts and maintenance tools required (see 3.27).
(p) Level of preservation and packaging and level of packing required (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the first article.

6.4 Standard lubricants. When requested by the contractor, the contracting officer should furnish a list of standard Military lubricants applicable to the equipment covered by this specification. Standard Military lubricants are listed in the DOD section of the Federal Supply Catalog, Department of Defense Section, Identification List, C9100-IL.

6.5 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:

Navy - YD

(Project No. 4130-N194)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-22014A(YD): CHILLER UNITS, LIQUID, PACKAGED, ELECTRIC MOTOR-DRIVEN WITH RECIPROCATING COMPRESSORS 10 TONS TO 150 TONS

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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